

AIE425 Intelligent Recommender Systems, Fall Semester 25/26

Final Course Project Plagiarism Report

Group 24

221100944 - Shiref Khaled Elhalawany

221100101– Ahmed Anis Hassan

221100391 - Karim Ashraf Elsayed

221101524– Kareem Shaheen

Submission Date: Monday, January 5, 2026

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Executive Summary:

This project delivers a complete, end-to-end study and implementation of recommender systems under **extreme sparsity**, using the **Amazon Reviews 2023 (Books, 5-core)** dataset ($\approx 9.49M$ interactions, **776k** users, **495k** items; sparsity $\approx 99.9975\%$). The work is organized into two complementary parts:

Section 1 (Matrix Factorization & Dimensionality Reduction) benchmarks **PCA-based latent modeling** (with both **mean-filled covariance** and **TRUE MLE covariance** under sparsity) against **SVD-based collaborative filtering** (full SVD and truncated SVD). It successfully constructs covariance structures, selects latent dimensionality via a **cross-validation rule**, and retrieves Top-5/Top-10 latent

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